



Friday Aug-31-2018

From:

Flowserve
1511 Jefferson St.

Sulphur Springs, TX, USA
75482
903-439-3358
kenhorne@flowserve.com

To:

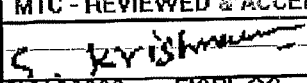
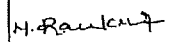
CERTIFICATE OF CONFORMANCE

Serial #	Heat # / Identifier	Customer PO #	Description	Class	End Connection	Shell Test	Seat Test
61-609450	0394S	STP46777	4" 2345	900	FLG	3375	2475
61-609451	0394S	STP46777	4" 2345	900	FLG	3375	2475
61-609452	0395S	STP46777	4" 2345	900	FLG	3375	2475
61-609453	0394S	STP46777	4" 2345	900	FLG	3375	2475

We hereby certify that all the subject valves/parts/sealant have been manufactured in conformance with the applicable requirements of Flowserve Sulphur Springs Operation (FSSO) specifications and that the valve/parts/sealant materials for component parts meet the applicable requirements of FSSO drawings and ASTM specifications, API Q1 9th edition, and API 6D 24th edition (if applicable to the API monogram). All valves/parts/sealant meet all the requirements of the customer purchase order.

Quality Engineer
Kenny Horne

[Handwritten signature]

	FLOW LINK SYSTEMS (P) LTD FOUNDRY UNIT - II 85/2 A,B & C, PUDUPALAYAM B.P.O AVINASHI TH, COIMBATORE - 641 554 INDIA		TEST CERTIFICATE (EN 10204-3.1)		TC NO : R-1665 DATE : 12-May-18									
	CUSTOMER FLOWSERVE INDIA CONTROLS PVT LTD, BB, MMDA Industrial Area, Maraimalai Nagar-603209 Kancheepuram, INDIA		SPECIFICATION REFERENCE 1. ASTM A216-16 Gr.WCC 2. Customer Spec. RMC 90012 Rev. 13		HEAT NO. 0394S Date of Pouring 26-Feb-18 FOUNDRY MARK FLS									
PO NUMBER & DATE 10081028 DT: 31.01.2018			ORDER ACCEPTANCE NO 18-0183											
CHEMICAL COMPOSITION % wt														
MELTING PRACTICE : Steel melted using Electric Induction furnace, and fully killed using Aluminium.														
SPECIFICATION	C	SI	S	P	Mn	NI	Cr	Mo	V	Cu	Al	-	*RE	**CE
Minimum	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Maximum	0.23	0.60	0.020	0.025	1.20	0.50	0.50	0.20	0.030	0.30	0.08	--	1.00	0.43
Achieved	0.20	0.43	0.006	0.015	0.97	0.07	0.08	0.009	0.000	0.01	0.05	--	0.17	0.38
Heat Treatment : NORMALIZING: Temperature raised to 920°C, Soaked for 4 hours and then atmospheric air-cooled														
Visual Inspection - Castings conforms to MSS-SP-55-2011 - Type II to XII (within level 'b')														
MECHANICAL PROPERTIES														
TEST TYPE	Tensile as per ASTM A370-17 / ISO 6892-4-2016						Impact (CVN) ISO 148-2016/ASTM A370-17		BEND TEST		HARDNESS			
SPECIFICATION	YS Rp 0.2% (KSI)	YS Rp 1.0% (KSI)	UTS (KSI)	% Elongation 4d 5d		% RA	Test Temperature (at °C) J		ANGLE	D	(HBW / HRBW)			
							Values	Avg.						
Minimum	40	--	70	22	--	35	--	--	--	--	--			
Maximum	--	--	95	--	--	--	--	--	--	--	--			
Achieved	46.1	--	75.6	40	--	68	--	--	--	--	153			
DETAIL OF CASTINGS														
S.No.	PO S.No.	DESCRIPTION		CAST DRAWING NO / REV		PART NO		POURED QTY		DESP. QTY				
1	02	4" 900 FE BODY		D-487876 / S		N2757928-50		6		6				
REMARKS: 1. Stated Chemical composition is from heat analysis; and Mechanical properties of test coupon from Cast test block. 2. *RE = Ni + Cr + Mo + V + Cu 3. ** CE = $C + \frac{Mn}{6} + \frac{Cr+Mo+V}{5} + \frac{Ni+Cu}{15}$ 4. The above stated KSI values have been calculated based on original tested NIMM ² values.														
CERTIFIED THAT 1. THE ABOVE GIVEN DETAILS ARE CORRECT, FULFILLS THE SPECIFICATION / STANDARDS AND PO. 2. MATERIAL WAS MANUFACTURED, SAMPLED, TESTED, INSPECTED IN ACCORDANCE WITH THE ABOVE SPECIFICATIONS AND CERTIFIED AS PER EN 10204 - 3.1. 3. THE MATERIAL IS FREE OF CONTAMINATION FROM RADIO ACTIVE ELEMENT OR RADIATION.						MTC - REVIEWED & ACCEPTED  10139966 FICPL QC  M.S.DHANABALAN ASSISTANT MANAGER - QUALITY MANUFACTURER'S AUTHORIZED INSPECTOR								
PREPARED BY: S.MANIBHARATHI			CHECKED BY:  M.RANJITH KUMAR											